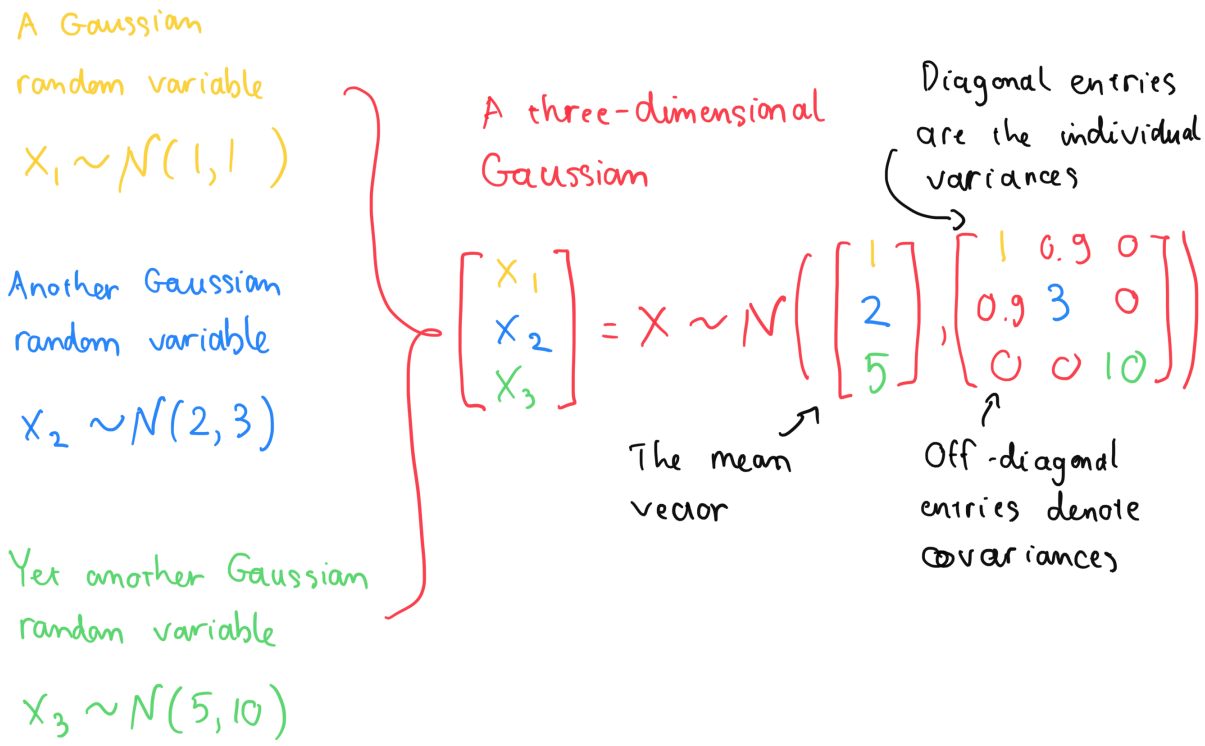


Figure 2.4 A multivariate Gaussian distribution groups multiple normally distributed random variables together. While the mean vector of the multivariate Gaussian concatenates the mean values, the covariance matrix models the correlations between the individual variables.

IMPORTANT **Definition of a multivariate Gaussian distribution**

Consider a random vector $X = [X_1\ X_2\ \ldots\ X_n]^T$ that follows a Gaussian distribution denoted by N (,), where is a vector of length n and is an n-by-n matrix. Here, is called the mean vector, whose individual elements denote the expected values of corresponding random variables in x, and is the covariance matrix, which describes the variance of individual variables as well as correlations between variables.

Let's take a moment to parse the above definition.

- First, due to the convenient properties of a multivariate Gaussian, each random variable in the vector x follows a normal distribution. Specifically, the i-th variable X_i has the mean value of $_i$, which is the i-th element of the mean vector of the multivariate Gaussian.
- Further, the variance of X_i is the i-th *diagonal* entry of the covariance matrix .
- If we have a vector of random variables following a multivariate Gaussian, then each individual variable corresponds a known normal distribution.

If the diagonal entries in the covariance matrix Σ are the variances of the individual variables, what about the off-diagonal entries? The entry in the i-th row and j-th column of this matrix denotes the covariance between X_i and X_j, which is related to the correlation between the two

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